Conditional expectation: from joint distributions to conditional averages

This bridge note supports learning outcome **S5**. It closes the notation and calculation gap between a joint PMF or PDF and conditional expectation; it does not replace the assigned Week 5 reading. Throughout, X and Y are real-valued random variables on a sample space Ω, while x and y denote their realised values. Write $\mathbb{P}(A)$ for the probability of an event A, $\mathbb{E}[X]$ for the expectation of X, and $\mathrm{Var}(X)$ for its variance.

1. Start with the joint support

For discrete X and Y, the joint PMF is

$$p_{X,Y}(x,y) := \mathbb{P}(X = x, Y = y).$$

Its (positive-probability) joint support is

$$S_{X,Y} := \{(x,y) : p_{X,Y}(x,y) > 0\}.$$

The marginal PMFs are obtained by summing over the other coordinate:

$$p_X(x) := \sum_y p_{X,Y}(x,y), \qquad p_Y(y) := \sum_x p_{X,Y}(x,y).$$

Write $S_X := \{x : p_X(x) > 0\}$ and $S_Y := \{y : p_Y(y) > 0\}$ for the marginal supports.

The sums run over the joint support (terms outside it are zero). A valid PMF is non-negative and sums to one; check these facts before conditioning. For a fixed y, the relevant conditional support is the slice

$$S_{X|Y=y} := \{x : (x,y) \in S_{X,Y}\}.$$

For continuous X and Y, let $f_{X,Y}(x,y)$ be their joint PDF and define the operational support

$$S_{X,Y} := \{(x,y) \in \mathbb{R}^2 : f_{X,Y}(x,y) > 0\}.$$

The marginal PDFs are the corresponding slices, written with integrals over $\mathbb{R}$:

$$f_X(x) := \int_{\mathbb{R}} f_{X,Y}(x,y)\,\mathrm{d}y, \qquad f_Y(y) := \int_{\mathbb{R}} f_{X,Y}(x,y)\,\mathrm{d}x.$$

When setting limits, use the inequalities that describe $S_{X,Y}$. For a fixed y, define $S_{X|Y=y} := \{x : (x,y) \in S_{X,Y}\}$. The density must be non-negative and integrate to one over its joint support.

2. Conditional PMFs and PDFs

Discrete case

If $p_Y(y) > 0$, the conditional PMF of X given $Y = y$ is

$$p_{X|Y}(x \mid y) := \mathbb{P}(X = x \mid Y = y) = \frac{p_{X,Y}(x,y)}{p_Y(y)}, \qquad x \in S_{X|Y=y}.$$

The denominator is the marginal probability of the conditioning value. A conditional PMF is not defined by this formula when $p_Y(y) = 0$; omit such a value rather than dividing by zero. For every admissible y, verify

$$\sum_{x \in S_{X|Y=y}} p_{X|Y}(x \mid y) = 1.$$

Continuous case

If $f_Y(y) > 0$, the conditional PDF is

$$f_{X|Y}(x \mid y) := \frac{f_{X,Y}(x, y)}{f_Y(y)}, \qquad x \in S_{X|Y=y}.$$

It is zero outside that conditional slice, and it integrates to one over the slice. The condition $f_Y(y) > 0$ is the density analogue of a positive conditioning probability. For values of y with $f_Y(y) = 0$, this ratio is not used. In a continuous model, the event $Y = y$ itself has probability zero, so the conditional PDF is the appropriate operational definition.

3. Two meanings of conditional expectation

Assume X is integrable, meaning

$$\mathbb{E}[|X|] < \infty.$$

For a fixed admissible y, the conditional expectation is a number. In the discrete case, the global condition above makes the following sum finite for every $y \in S_Y$:

$$m(y) := \mathbb{E}[X \mid Y = y] := \sum_{x \in S_{X|Y=y}} x\, p_{X|Y}(x \mid y),$$

In the continuous case, the displayed integral defines $m(y)$ for f_Y-almost every y with $f_Y(y) > 0$:

$$m(y) := \mathbb{E}[X \mid Y = y] := \int_{S_{X|Y=y}} x\, f_{X|Y}(x \mid y)\, \mathrm{d}x.$$

Set $m(y) := 0$ on the exceptional set if a value is needed. The particular choice there does not affect any probability or expectation because that set has zero probability under Y.

The notation $m(y)$ emphasises that the value depends on the fixed number y. The conditional-expectation random variable is instead

$$\mathbb{E}[X \mid Y] := m(Y) \quad \text{almost surely.}$$

For a discrete Y, it takes the value $m(y)$ when $Y = y$. Thus $\mathbb{E}[X \mid Y = y]$ is a scalar indexed by y, whereas $\mathbb{E}[X \mid Y]$ is a random variable whose randomness comes from Y. Do not report one in place of the other.

Conditioning on an event

Let $B \subseteq \Omega$ be an event with $\mathbb{P}(B) > 0$. For $\omega \in \Omega$, define its indicator by

$$\mathbb{1}_B(\omega) := \begin{cases} 1, & \omega \in B, \\ 0, & \omega \notin B. \end{cases}$$

For integrable X, conditioning on B gives the scalar

$$\mathbb{E}[X \mid B] := \frac{\mathbb{E}[X\mathbb{1}_B]}{\mathbb{P}(B)}.$$

In the discrete case $B = \{Y = y\}$ reproduces $\mathbb{E}[X \mid Y = y]$ when $p_Y(y) > 0$. If $B_1, \ldots, B_k$ are disjoint events that partition Ω, each with positive probability, then the event version of total expectation is

$$\mathbb{E}[X] = \sum_{i=1}^{k} \mathbb{E}[X \mid B_i]\, \mathbb{P}(B_i).$$

4. Independence: what follows, and what does not

The factorisation test is exact. Discrete X and Y are independent if

$$p_{X,Y}(x, y) = p_X(x)p_Y(y) \quad \text{for every } x, y,$$

and jointly continuous X and Y are independent if

$$f_{X,Y}(x, y) = f_X(x)f_Y(y) \quad \text{for two-dimensional Lebesgue-a.e. } (x, y),$$

that is, outside a set of two-dimensional Lebesgue measure zero. Equivalently, in the discrete case $p_{X|Y}(x \mid y) = p_X(x)$ for every x and every y with $p_Y(y) > 0$; in the continuous case,

$$f_{X|Y}(x \mid y) = f_X(x) \quad \text{for one-dimensional Lebesgue-a.e. } x, \quad \text{for } f_Y\text{-a.e. } y \text{ with } f_Y(y) > 0.$$

Checking one cell or one conditioning value is not enough to establish independence.

If X and Y are independent and X is integrable, then

$$\mathbb{E}[X \mid Y = y] = \mathbb{E}[X] \quad \text{and} \quad \mathbb{E}[X \mid Y] = \mathbb{E}[X]$$

for all admissible conditioning values (or for f_Y-a.e. such value in the continuous case). The converse is weaker: a constant conditional mean only says that X is mean-independent of Y, not that their full conditional distribution factorises. Likewise, zero covariance, when defined, does not by itself establish independence.

5. Iterated expectation and total variance

Law of iterated expectation

For integrable X, the law of iterated expectation is

$$\boxed{\mathbb{E}[\mathbb{E}[X \mid Y]] = \mathbb{E}[X].}$$

For a discrete Y, this means

$$\mathbb{E}[X] = \sum_{y \in S_Y} m(y)p_Y(y),$$

and for a continuous Y,

$$\mathbb{E}[X] = \int_{\mathbb{R}} m(y) f_Y(y)\,\mathrm{d}y.$$

With $\mathbb{E}[|X|] < \infty$, this integral is well-defined for the version of m above: changing m on an f_Y-null set does not change it, and

$$\int m(y) f_Y(y)\,\mathrm{d}y = \iint x f_{X|Y}(x \mid y) f_Y(y)\,\mathrm{d}x\,\mathrm{d}y = \iint x f_{X,Y}(x,y)\,\mathrm{d}x\,\mathrm{d}y = \mathbb{E}[X].$$

The weights are the marginal distribution of the conditioning variable, not equal weights unless that happens to be the model.

Law of total variance

Now assume the finite-second-moment condition

$$\mathbb{E}[X^2] < \infty.$$

In the discrete case, the global second-moment condition makes the following conditional variance finite for every $y \in S_Y$:

$$v(y) := \mathrm{Var}(X \mid Y = y) := \sum_{x \in S_{X|Y=y}} (x - m(y))^2 p_{X|Y}(x \mid y),$$

In the continuous case, define

$$v(y) := \int_{S_{X|Y=y}} (x - m(y))^2 f_{X|Y}(x \mid y)\,\mathrm{d}x$$

for f_Y-almost every y with $f_Y(y) > 0$, and set $v(y) := 0$ elsewhere. Then

$$\mathrm{Var}(X \mid Y) := v(Y) \quad \text{almost surely.}$$

Let $\mu := \mathbb{E}[X]$. Since $\mathbb{E}[X \mid Y] = m(Y)$, the decomposition

$$X - \mu = (X - m(Y)) + (m(Y) - \mu)$$

has a conditional cross-term with expectation zero. Taking conditional and then unconditional expectations gives

$$\begin{aligned} \mathrm{Var}(X) &= \mathbb{E}[\mathbb{E}[(X-\mu)^2 \mid Y]] \\ &= \mathbb{E}\Big[v(Y) + (m(Y)-\mu)^2\Big] \\ &= \boxed{\mathbb{E}[\mathrm{Var}(X \mid Y)] + \mathrm{Var}(\mathbb{E}[X \mid Y])\,.} \end{aligned}$$

The first term is average variation within the conditional groups. The second term is variation between their conditional means. Both terms are needed.

6. Worked example — one table, all the objects

Let $X \in \{0,1\}$ and $Y \in \{0,1\}$ have the joint PMF below. All four cells are in the joint support.

$x\backslash y$	0	1
0	1/4	1/8
1	1/4	3/8

The marginals are

$$p_Y(0) = p_Y(1) = \frac{1}{2}, \qquad p_X(0) = \frac{3}{8}, \qquad p_X(1) = \frac{5}{8}.$$

For $Y = 0$, divide the first column by $p_Y(0) = 1/2$; for $Y = 1$, divide the second column by $p_Y(1) = 1/2$. Thus

y	$p_{X\mid Y}(0 \mid y)$	$p_{X\mid Y}(1 \mid y)$	$m(y) = \mathbb{E}[X \mid Y = y]$
0	1/2	1/2	1/2
1	1/4	3/4	3/4

The conditional-expectation random variable is therefore

$$\mathbb{E}[X \mid Y] = \begin{cases} 1/2, & Y = 0, \\ 3/4, & Y = 1. \end{cases}$$

The variables are not independent because, for example,

$$p_{X,Y}(1,1) = \frac{3}{8} \neq p_X(1)p_Y(1) = \frac{5}{8} \cdot \frac{1}{2} = \frac{5}{16}.$$

Iterated expectation gives

$$\mathbb{E}[\mathbb{E}[X \mid Y]] = \frac{1}{2} \cdot \frac{1}{2} + \frac{3}{4} \cdot \frac{1}{2} = \frac{5}{8},$$

which agrees with $\mathbb{E}[X] = p_X(1) = 5/8$. Conditioning on the event $B = \{Y = 1\}$ gives the same scalar, $\mathbb{E}[X \mid B] = 3/4$, because $\mathbb{P}(B) = 1/2$.

For total variance, the conditional Bernoulli variances are

$$v(0) = \frac{1}{2}\frac{1}{2} = \frac{1}{4}, \qquad v(1) = \frac{3}{4}\frac{1}{4} = \frac{3}{16}.$$

Hence

$$\mathbb{E}[\mathrm{Var}(X \mid Y)] = \frac{1}{2} \cdot \frac{1}{4} + \frac{1}{2} \cdot \frac{3}{16} = \frac{7}{32},$$

while

$$\mathrm{Var}(\mathbb{E}[X \mid Y]) = \left(\frac{1}{2} \cdot \frac{1}{4} + \frac{1}{2} \cdot \frac{9}{16}\right) - \left(\frac{5}{8}\right)^2 = \frac{1}{64}.$$

The total is $7/32 + 1/64 = 15/64$. Directly, X is marginally Bernoulli(5/8), so

$$\mathrm{Var}(X) = \frac{5}{8}\frac{3}{8} = \frac{15}{64},$$

which verifies the decomposition.

7. Pitfalls and final check

- Write the joint support first. Marginalise over the other variable, and carry the resulting conditional slice into every sum or integral.
- Divide by $p_Y(y)$ or $f_Y(y)$, not by the wrong marginal and never by a zero denominator.
- Keep $\mathbb{E}[X \mid Y = y]$ (a number for fixed y) separate from $\mathbb{E}[X \mid Y]$ (the random variable obtained by substituting Y).
- In iterated expectation and total variance, weight conditional quantities by the marginal distribution of Y.
- Total variance has a within-group term and a between-means term; dropping either changes the answer.
- Independence implies a constant conditional distribution and conditional mean. A single matching cell, a constant conditional mean or zero covariance is not a converse proof of independence.
- Before using an expectation, check $\mathbb{E}[|X|] < \infty$; before using a variance decomposition, check $\mathbb{E}[X^2] < \infty$.

Read the assigned sections of Pishro-Nik, §§5.1.1–5.1.3 and 5.1.5 on joint, marginal, conditional and expected distributions and §§5.2.1–5.2.3 on joint and conditional PDFs, following the Week 5 reading guide's skips. This note is an original institutional explanation; it does not reproduce, adapt or replace that source.